

3) Basic properties of quantum state

a) $\otimes B^+ = \left[\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right]^+ = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = B$

b) Purity
 $B = \begin{pmatrix} \frac{3}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{1}{4} \end{pmatrix}$

$$S^2 = \begin{vmatrix} \frac{3}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{1}{4} \end{vmatrix} = \begin{vmatrix} \frac{9}{16} & \frac{4\sqrt{3}}{16} \\ \frac{4\sqrt{3}}{16} & \frac{1}{16} \end{vmatrix}$$

$B^+ = B$

Purity = $\text{Tr}[S^2]$

$$\text{Tr}[S^2] = \frac{12}{16} + \frac{4}{16} = \frac{16}{16}$$

$$= \frac{16}{16} = 1$$

so it is pure state

④ Trace (Tr)

$$\begin{aligned} \text{Tr}[B] &= \text{Tr}\left[\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right] \\ &= \frac{1}{2} + \frac{1}{2} = 1 \\ &= 1 \end{aligned}$$

c) Given

$$| \psi \rangle = \frac{1}{2} [| 00 \rangle + | 01 \rangle + | 10 \rangle - | 11 \rangle]$$

Solution

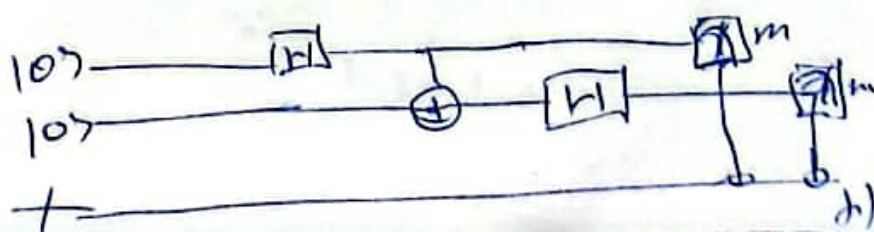
$$| \psi \rangle = \frac{1}{2} \left[| 0 \rangle \left[\frac{| 0 \rangle + | 1 \rangle}{\sqrt{2}} \right] + \left[\frac{| 0 \rangle - | 1 \rangle}{\sqrt{2}} \right] | 1 \rangle \right]$$

$$| \psi \rangle = \frac{1}{2} \left[| 0 \rangle \left[\frac{| 0 \rangle + | 1 \rangle}{\sqrt{2}} \right] + \left[\frac{| 0 \rangle - | 1 \rangle}{\sqrt{2}} \right] | 1 \rangle \right]$$

Schmidt Rank = 2

d) Given $| \psi \rangle = \frac{1}{2} [| 00 \rangle + | 10 \rangle + | 01 \rangle - | 11 \rangle]$

Quantum circuit



3) Positive Eigenvalue ($\lambda \geq 0$)

$$\det[B - \lambda I] = 0$$

$$\det\left[\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right] = 0$$

$$\det\left[\begin{bmatrix} \frac{1}{2} - \lambda & 0 \\ 0 & \frac{1}{2} - \lambda \end{bmatrix}\right] = 0$$

$$\left(\frac{1}{2} - \lambda\right)\left(\frac{1}{2} - \lambda\right) - \frac{0}{4} = 0$$

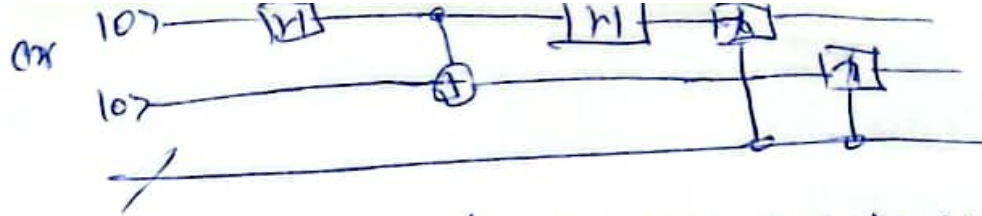
$$\left(\frac{1}{2} - \lambda\right)^2 = \frac{1}{4}$$

$$\left(\frac{1}{2} - \lambda\right) = \pm \frac{1}{2}$$

$$\lambda = \frac{1}{2} \pm \frac{1}{2}$$

$$\lambda_1 = \frac{1}{2} + \frac{1}{2}, \lambda_2 = \frac{1}{2} - \frac{1}{2}$$

$$\lambda_1 = 1, \lambda_2 = 0$$



both give, $147 = \frac{(1007 + 107 + 1707 - 1777)}{2}$

e) Solution

$$147 = \frac{1}{2} [1007 + 1707 + 107 - 1777]$$

Apply ~~the~~

$$\begin{aligned} 147 &= \frac{1}{2} [(1007 + 1707) + (107 - 1777)] \\ &= \frac{1}{2} \left[\frac{(107 + 177)}{12} 1007 + \left(\frac{107 - 177}{12} \right) 177 \right] \\ &= \frac{1}{2} [1+07 + 1-77] \\ &= \underline{\underline{11}} \end{aligned}$$

Curia Rank $(r) = 2$, where, $r \geq 2$, is Entangled

3) Depolarizing Channel Example

0) Solution

$$I = 2^n \times 2^n$$

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{2^n} I$$

Apply Tr

$$\begin{aligned} \text{Tr}[\mathcal{E}(\rho)] &= \text{Tr} \left[(1-p)\rho + \frac{p}{2^n} I \right] = (1-p) \text{Tr}[\rho] + \frac{p}{2^n} \text{Tr}[I] \\ &= (1-p) \text{Tr}[\rho] + \frac{p}{2^n} \text{Tr}[I] \\ &= (1-p) \cdot 1 + \frac{p}{2^n} \cdot 2^n \\ &= 1-p + p \\ &= \underline{\underline{1}} \end{aligned}$$

b) Solution

$$Z(S) = (1-p)S + \frac{p}{2}I, \text{ for } n=2$$

$$M_0 = \sqrt{1 - \frac{3p}{4}}, M_1 = \frac{\sqrt{p}}{2}X, M_2 = \frac{\sqrt{p}}{2}Y, M_3 = \frac{\sqrt{p}}{2}Z$$

$$\Sigma(S) = \sum_{k=0}^3 M_k^+ S M_k$$

$$= \left(\sqrt{1 - \frac{3p}{4}} \right)^+ S \left(\sqrt{1 - \frac{3p}{4}} \right) + \left(\frac{\sqrt{p}}{2}X \right)^+ S \left(\frac{\sqrt{p}}{2}X \right) + \left(\frac{\sqrt{p}}{2}Y \right)^+ S \left(\frac{\sqrt{p}}{2}Y \right) + \left(\frac{\sqrt{p}}{2}Z \right)^+ S \left(\frac{\sqrt{p}}{2}Z \right)$$

$$= \left(1 - \frac{3p}{4} \right) S + \frac{p}{4} X^+ S X + \frac{p}{4} Y^+ S Y + \frac{p}{4} Z^+ S Z$$

$$= \left(1 - \frac{3p}{4} \right) S + \frac{p}{4} \underbrace{[X^+ S X + Y^+ S Y + Z^+ S Z]}_{= 2I - S}$$

$$= \left(1 - \frac{3p}{4} \right) S + \frac{p}{4} [2I - S]$$

$$= S - \frac{3p}{4}S + p\frac{I}{2} - p\frac{S}{4} = S - \frac{4p}{4}S + \frac{1}{2}pI$$

$$= S - pS + \frac{1}{2}pI = (1-p)S + p\frac{I}{2}$$

$$= (1-p)S + p\frac{I}{2}$$



c) Solution

$$Z(S) = (1-p)S + \frac{p}{2}I$$

Calculate P_{n+1}

$$P_{n+1} = \text{Tr}[Z^2(S)]$$

$$\text{Tr}[Z^2(S)] = \text{Tr} \left[(1-p)S + p\frac{I}{2} \right] \left[(1-p)S + p\frac{I}{2} \right]$$

$$= \text{Tr} \left[(1-p)^2 S^2 + (1-p)p\frac{I}{2}S + p\frac{I}{2}(1-p)S + \frac{p^2}{4}I^2 \right]$$

$$= (1-p)^2 \text{Tr}[S^2] + 2p\frac{I}{2}(1-p) \text{Tr}[S] + \frac{p^2}{4} \text{Tr}[I^2]$$

3)

$$= (\lambda - p)^2 \text{Tr} \left[\frac{1}{4} (I + 2\mathbf{r} \cdot \boldsymbol{\sigma} + |\mathbf{r}|^2 I) \right] + p(\lambda - p) \cdot 1 + \frac{p^2}{4} \cdot 2$$

$$= (\lambda - p)^2 \text{Tr} \left[\frac{1}{4} (I + 2\mathbf{r} \cdot \boldsymbol{\sigma} + |\mathbf{r}|^2 I) \right] + p(\lambda - p) + \frac{p^2}{2}$$

$$= \frac{(\lambda - p)^2}{4} [\text{Tr}(I) + 2\mathbf{r} \cdot \text{Tr}(\boldsymbol{\sigma}) + |\mathbf{r}|^2 \text{Tr}(I)] + p(\lambda - p) + \frac{p^2}{2}$$

$$= \frac{(\lambda - p)^2}{4} [2 + 2\mathbf{r} \cdot 0 + 2|\mathbf{r}|^2] + p(\lambda - p) + \frac{p^2}{2}$$

$$= \frac{(\lambda - p)^2}{4} [2 + 2|\mathbf{r}|^2] + p(\lambda - p) + \frac{p^2}{2}$$

$$= \frac{2}{4} (\lambda - p)^2 [1 + |\mathbf{r}|^2] + p(\lambda - p) + \frac{p^2}{2}$$

$$= \frac{1}{2} (\lambda - p)^2 [1 + |\mathbf{r}|^2] + p - p^2 + \frac{p^2}{2}$$

$$= \frac{(\lambda - 2p + p^2)}{2} + \frac{(\lambda - p)^2 |\mathbf{r}|^2}{2} + p - p^2 + \frac{p^2}{2}$$

$$= \frac{(\lambda - p)^2 |\mathbf{r}|^2}{2} + \frac{1}{2} - \frac{2p}{2} + \frac{p^2}{2} + p - p^2 + \frac{p^2}{2}$$

$$= \frac{(\lambda - p)^2 |\mathbf{r}|^2}{2} + \underbrace{\frac{1}{2} - \frac{2p}{2} + p + \frac{p^2}{2} - p^2 + \frac{p^2}{2}}_{=0}$$

$$= \frac{1 + (\lambda - p)^2 |\mathbf{r}|^2}{2} + 0$$

$$= \frac{1 + (\lambda - p)^2 |\mathbf{r}|^2}{2} \quad \#$$

Block vectors scales as

$$|\mathbf{r}| \rightarrow p(\lambda - p) |\mathbf{r}|$$

- Purify is unchanged only if $|\mathbf{r}| = 0$ (maximum mixed) or $p = 1$

(4)

d) Solution

$$\rho_{AB} = |\psi\rangle\langle\psi|$$

$$= \frac{1}{2} [|\psi\rangle\langle\psi| + |\psi\rangle\langle\psi| + |\psi\rangle\langle\psi| + |\psi\rangle\langle\psi|]$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Calculate partial transpose

$$\rho_{AB}^{TB} =$$

$$\rho_{AB}^{TB} = [|\psi\rangle\langle\psi|]^{TB} = \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Calculate the partial transpose of the state $\rho_{AB} = \frac{1}{2}(|\psi\rangle\langle\psi| + |\psi\rangle\langle\psi|)$

$$\rho_{AB}^{TB} = (2-p) \rho_{AB} + \frac{p}{4} I_4$$

$$= \left(\frac{2-p}{2}\right) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \frac{p}{4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2-p}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & \frac{2-p}{2} & 0 \\ 0 & \frac{2-p}{2} & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{2-p}{4} \end{bmatrix}$$

e) $\rho_{AB} = \frac{1}{2}(|\psi\rangle\langle\psi| + |\psi\rangle\langle\psi|)$. What is the value of p - Entangled?

$$\text{For } \rho_{AB} = (2-p) \rho_{AB} + \frac{p}{4} I_4$$

To detect the state, separable and entangled

Calculate the fidelity

For - separable F&F fidelity with $|\phi^+\rangle$, $F \leq \frac{1}{2}$ }
- Entangled iff the fidelity with $|\phi^+\rangle$, $F > \frac{1}{2}$ }

Calculate,

$$F = \langle \phi^+ | \rho | \phi^+ \rangle - \text{for pure state,}$$

$$= \langle \phi^+ | \left((2-p) |\phi^+\rangle \langle \phi^+| + \frac{p}{4} \right) | \phi^+ \rangle$$

$$= (2-p) \underbrace{\langle \phi^+ | \phi^+ \rangle}_{=1} \underbrace{\langle \phi^+ | \phi^+ \rangle}_{=1} + \frac{p}{4} \underbrace{\langle \phi^+ | \phi^+ \rangle}_{=1}$$

$$= (2-p) + \frac{p}{4}$$

Therefore

$$(2-p) + \frac{p}{4} > \frac{1}{2}$$

$$(2-p) + \frac{p}{4} > \frac{1}{2}$$

$$2 - p + \frac{p}{4} > \frac{1}{2}$$

$$2 - \frac{4p + p}{4} > \frac{1}{2}$$

$$2 - \frac{5p}{4} > \frac{1}{2}$$

$$-\frac{5p}{4} > \frac{1}{2} - 2$$

$$-\frac{5p}{4} > -\frac{3}{2}$$

$$\frac{5p}{4} < \frac{3}{2}$$

$$\frac{5p}{2} < 3$$

$$p < \frac{2}{3}$$

Therefore

$$p < \frac{2}{3} - \text{for}$$

Entangled state.

Entangled F&F

So Entangled for $0 \leq p < \frac{2}{3}$

and separable for $p \geq \frac{2}{3}$

$0 \leq p < \frac{2}{3} \rightarrow$ Entangled }
 $p \geq \frac{2}{3} \rightarrow$ Separable }

2) Bell inequality

Given

$$\langle A_0 B_0 \rangle + \langle A_0 B_2 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_2 \rangle \leq 2$$

$$\begin{aligned} n_0 &= (\cos(\theta), 0, \sin(\theta)) \\ &= (\cos(\frac{\pi}{4}), 0, \sin(\frac{\pi}{4})) \\ &= (\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}) \end{aligned}$$

$$\begin{aligned} n_2 &= (\sin(\theta), 0, -\cos(\theta)) \\ &= (\sin(\frac{\pi}{4}), 0, -\cos(\frac{\pi}{4})) \\ &= (\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}) \end{aligned}$$

$$A_0 = m_0 \cdot \hat{a}$$

$$= (2, 0, 0) \cdot (x, y, z)$$

$$= \underline{\underline{x}}$$

$$A_1 = m_1 \cdot \hat{a}$$

$$= (0, 0, 2) \cdot (x, y, z)$$

$$= \underline{\underline{2z}}$$

$$B_0 = n_0 \cdot \hat{b}$$

$$= (\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}) \cdot (x, y, z)$$

$$= \underline{\underline{\frac{\sqrt{2}}{2}(x+z)}}$$

$$B_1 = n_1 \cdot \hat{b}$$

$$= (\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}) \cdot (x, y, z)$$

$$= \underline{\underline{\frac{\sqrt{2}}{2}(x-z)}}$$

(a) Solution

$$\langle A_0 B_0 \rangle + \langle A_0 B_2 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_2 \rangle \leq 2$$

$$x(\frac{\sqrt{2}}{2}(x+z)) + x(\frac{\sqrt{2}}{2}(x-z)) + 2(\frac{\sqrt{2}}{2}(x+z)) - 2(\frac{\sqrt{2}}{2}(x-z)) \leq 2$$

$$\frac{\sqrt{2}}{2} (xx + \cancel{xz} + \cancel{xx} - \cancel{xz} + 2x + 2z - 2x + 2z) \leq 2$$

$$\frac{\sqrt{2}}{2} (xx + \cancel{xx} + 2z + 2z) \leq 2$$

$$\frac{\sqrt{2}}{2} (2xx + 2zz) \leq 2$$

$$\frac{\sqrt{2}}{2} (x(x+z) + z(x+z)) \leq 2$$

$$xx + zz \leq \frac{2}{\sqrt{2}}$$

$$xx + zz \leq \frac{2 \times \sqrt{2}}{\sqrt{2}}$$

$$xx + zz \leq \frac{2}{1} \sqrt{2}$$

$$\underline{\underline{xx + zz \leq 2}}$$

b) Bfuer

$$|\psi\rangle = \sqrt{\lambda-E} |00\rangle + \sqrt{E} |11\rangle$$

Squation

$$\langle XX \rangle = \langle \psi | XX | \psi \rangle$$

$$\begin{aligned} &= (\sqrt{\lambda-E} \langle 00| + \sqrt{E} \langle 11|) XX (\sqrt{\lambda-E} |00\rangle + \sqrt{E} |11\rangle) \\ &= (\sqrt{\lambda-E} \langle 00| + \sqrt{E} \langle 11|) (\sqrt{\lambda-E} XX |00\rangle + \sqrt{E} XX |11\rangle) \\ &= (\sqrt{\lambda-E} \langle 00| + \sqrt{E} \langle 11|) (\sqrt{\lambda-E} |11\rangle + \sqrt{E} |00\rangle) \\ &= \sqrt{\lambda-E} \sqrt{E} + \sqrt{E} \sqrt{\lambda-E} = 2\sqrt{E(\lambda-E)} \\ &= \underline{\underline{2\sqrt{E(\lambda-E)}}} \end{aligned}$$

$$\langle ZZ \rangle = \langle \psi | ZZ | \psi \rangle$$

$$\begin{aligned} &= (\sqrt{\lambda-E} \langle 00| + \sqrt{E} \langle 11|) (ZZ) (\sqrt{\lambda-E} |00\rangle + \sqrt{E} |11\rangle) \\ &= (\sqrt{\lambda-E} \langle 00| + \sqrt{E} \langle 11|) (\sqrt{\lambda-E} |00\rangle + \sqrt{E} |11\rangle) \\ &= \lambda - E + E \\ &= \underline{\underline{\lambda}} \end{aligned}$$

Which values of E is CHSH inequality violated?

$$\langle XX \rangle + \langle ZZ \rangle > \sqrt{2}$$

$$2\sqrt{E(\lambda-E)} + \lambda > \sqrt{2}$$

$$\frac{2\sqrt{E(\lambda-E)}}{\lambda} > \frac{\sqrt{2}-\lambda}{\lambda}$$

$$\sqrt{E(\lambda-E)} > \frac{\sqrt{2}-1}{2}$$

$$(\sqrt{E(1-E)})^2 > \left(\frac{\sqrt{2}-1}{2}\right)^2$$

$$E(1-E) > \frac{3-2\sqrt{2}}{4}$$

$$E - E^2 > \frac{3-2\sqrt{2}}{4}$$

$$E^2 - E + \frac{3-2\sqrt{2}}{4} < 0$$

$$E = \frac{1 \pm \sqrt{1 - 4 \cdot \frac{3-2\sqrt{2}}{4}}}{2}$$

$$= \frac{1 \pm \sqrt{2(\sqrt{2}-1)}}{2}$$

$$\frac{1 - \sqrt{2(\sqrt{2}-1)}}{2} < E < \frac{1 + \sqrt{2(\sqrt{2}-1)}}{2}$$

~~← / / →~~

c) Solution

$$S(E) = 4XX + 4ZZ$$

$$S(E) = 2 + 2\sqrt{E(1-E)}$$

Then, $S(E)$ is maximum at $E = \frac{1}{2}$

$$S\left(\frac{1}{2}\right) = 2 + 2\sqrt{\frac{1}{2}\left(1 - \frac{1}{2}\right)}$$

$$= 2 + 2\sqrt{\frac{1}{2}\left(\frac{2-1}{2}\right)}$$

$$= 2 + 2\sqrt{\frac{1}{2} \cdot \frac{1}{2}}$$

$$= 2 + 2 \cdot \frac{1}{2}$$

$$= 2 + 1$$

$$= \underline{\underline{3}}$$

Maximum occurs at

$$E = \frac{1}{2} \quad \text{?}$$

$$S_{\max} = 3$$

d) Solution

$$\text{Energy levels: } 4XX + 4ZZ \leq \sqrt{2}$$

$$1\phi = \frac{100 - 111}{\sqrt{2}}$$

$$4XX = 4\phi - 111$$

$$= \frac{1}{2}(100 - 111) \frac{1}{\sqrt{2}}(111 - 100)$$

$$= \frac{1}{2}(100 - 111)(111 - 100)$$

$$= \frac{1}{2}(-1 - 1)$$

$$= -\frac{2}{2}$$

$$= \underline{\underline{-1}}$$

$$4ZZ = 4\phi - 111$$

$$= \frac{1}{2}(100 - 111)(111 - 100)$$

$$= \frac{1}{2}(100 - 111)(111 - 100)$$

$$= \frac{1}{2}(111 - 100)$$

$$= \frac{1}{2}$$

$$= \underline{\underline{1}}$$

$$S(E) = 4XX + 4ZZ$$

$$= -1 + 1$$

$$= \underline{\underline{0}}$$

$$\underline{\underline{0 \leq 2}}$$

e) : Solution

$$|P\rangle = |a\rangle \otimes |b\rangle$$

$$\begin{aligned} \langle XX \rangle &= \langle X \otimes X \rangle \\ &= \langle X \rangle_a \langle X \rangle_b \end{aligned}$$

$$\begin{aligned} \langle ZZ \rangle &= \langle Z \otimes Z \rangle \\ &= \langle Z \rangle_a \langle Z \rangle_b \end{aligned}$$

$$\left. \begin{aligned} a_1 &= \langle X \rangle_a \\ a_2 &= \langle Z \rangle_a \\ b_1 &= \langle X \rangle_b \\ b_2 &= \langle Z \rangle_b \end{aligned} \right\}$$

$$\begin{aligned} \langle XX \rangle + \langle ZZ \rangle &= \langle X \rangle_a \langle X \rangle_b + \langle Z \rangle_a \langle Z \rangle_b \\ &= a_1 b_1 + a_2 b_2 \end{aligned}$$

Use Cauchy-Schwarz

The 2D Vectors (a_1, a_2) and (b_1, b_2) . Then

$$a_1 b_1 + a_2 b_2 \leq \sqrt{a_1^2 + a_2^2} \sqrt{b_1^2 + b_2^2}$$

Bound each length by 1

For any single-qubit state, the Bloch vector has length ≤ 1 . In particular

$$\left. \begin{aligned} a_1^2 + a_2^2 &\leq 1 \\ b_1^2 + b_2^2 &\leq 1 \end{aligned} \right\}$$

$$\begin{aligned} a_1 b_1 + a_2 b_2 &\leq \sqrt{a_1^2 + a_2^2} \sqrt{b_1^2 + b_2^2} \\ &\leq \sqrt{1} \sqrt{1} \\ &\leq 1 \cdot 1 \end{aligned}$$

$$\begin{aligned} a_1 b_1 + a_2 b_2 &\leq 1 \\ \Rightarrow \langle XX \rangle + \langle ZZ \rangle &\leq 1 \end{aligned}$$

$$\langle XX \rangle + \langle ZZ \rangle \leq 1$$

f) Solution

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|$$

Show that

$$\langle XX \rangle + \langle ZZ \rangle \leq 1$$

Expectation Values are linear in the state

$$\langle O \rangle_\rho = \text{Tr}(\rho O), \quad \text{where } O = XX + ZZ$$

$$\langle XX \rangle_\rho + \langle ZZ \rangle_\rho = \text{Tr}(\rho(XX + ZZ)) = \text{Tr}(\rho O)$$

$$\begin{aligned} \langle XX \rangle_\rho + \langle ZZ \rangle_\rho &= \text{Tr}\left(\sum_k p_k |\psi_k\rangle\langle\psi_k| O\right) \\ &= \sum_k p_k \text{Tr}(|\psi_k\rangle\langle\psi_k| O) \\ &= \sum_k p_k \langle O \rangle_{|\psi_k\rangle} = \langle O \rangle_{|\psi_k\rangle} \\ &\quad \underline{\underline{1}} \end{aligned}$$

$$\begin{aligned} \langle O \rangle_{|\psi_k\rangle} &= \langle XX \rangle_{|\psi_k\rangle} + \langle ZZ \rangle_{|\psi_k\rangle} \leq 1 \\ &= \underline{\underline{1}} \end{aligned}$$

So

$$\langle O \rangle_\rho = \sum_k p_k \langle O \rangle_{|\psi_k\rangle} \leq \sum_k p_k \cdot 1$$

$$\leq \sum_k p_k = 1$$

$$\leq 1$$

$$\langle O \rangle_\rho = \sum_k p_k \langle O \rangle_{|\psi_k\rangle} \leq 1$$

$$\langle XX \rangle + \langle ZZ \rangle \leq 1$$

$\langle XX \rangle + \langle ZZ \rangle \leq 1$, for all separable states,

4 Rydberg Blockade

a) Given

$$H_A = \frac{\Omega}{2} \sigma_x + \frac{\Delta}{2} \sigma_z, \quad | \psi(0) \rangle = | 0 \rangle$$

Solution

Rabi frequency

$$\Omega_R = \sqrt{\Omega^2 + \Delta^2}$$

The excited-state population is

$$P_{|1\rangle}(t) = \frac{\Omega^2}{\Omega^2 + \Delta^2} \sin^2\left(\frac{\Omega_R t}{2}\right)$$

$$= \frac{\Omega^2}{\Omega_R^2} \sin^2\left(\frac{\Omega_R t}{2}\right)$$

The maximal population,

$$\text{at } \sin^2\left(\frac{\Omega_R t}{2}\right) = 1,$$

$$P_{|1\rangle}(t) = \frac{\Omega^2}{\Omega^2 + \Delta^2} \cdot 1 = \frac{\Omega^2}{\Omega^2 + \Delta^2}$$

In the limit $\frac{\Omega}{\Delta} \ll 1$

$$P_{\max} \approx \left(\frac{\Omega}{\Delta}\right)^2, \quad (\text{strongly suppressed})$$

b) Solution

$$H_S = \frac{\Omega}{2} \sum_{i=2}^3 \sigma_x^{(i)} + V \sum_{i < j} |1\rangle_i \langle 1|_j$$

For,
 $n = 3$

Number of states $= 2^n = 2^3 = 8$

$$\left\{ \begin{array}{ll} |000\rangle & |100\rangle \\ |001\rangle & |101\rangle \\ |010\rangle & |110\rangle \\ |011\rangle & |111\rangle \end{array} \right\}$$

Diagonal (Interaction) terms:

$$V = \langle 011 | H_3 | 011 \rangle$$

$$V = \langle 101 | H_3 | 101 \rangle$$

$$V = \langle 110 | H_3 | 110 \rangle$$

$$3V = \langle 111 | H_3 | 111 \rangle$$

All other diagonal elements are 0.

Off-diagonal (drive) terms: $\frac{\Omega}{2}$

between any two basis states that differ by flipping exactly one qubit

Explicitly, the non-zero couplings are:

$$|000\rangle \leftrightarrow |001\rangle, |010\rangle, |100\rangle$$

$$|001\rangle \leftrightarrow |011\rangle, |101\rangle$$

$$|010\rangle \leftrightarrow |011\rangle, |110\rangle$$

$$|100\rangle \leftrightarrow |101\rangle, |110\rangle$$

$$|011\rangle \leftrightarrow |111\rangle$$

$$|101\rangle \leftrightarrow |111\rangle$$

Note, that: listed pairs are matrix elements

$$\langle a | H_3 | b \rangle = \langle b | H_3 | a \rangle = \frac{\Omega}{2}$$



The full 8x8 matrix

All these are symmetric

$$H_3 = \begin{pmatrix} 0 & \frac{\Omega}{2} & \frac{\Omega}{2} & 0 & \frac{\Omega}{2} & 0 & 0 & 0 \\ \frac{\Omega}{2} & 0 & 0 & \frac{\Omega}{2} & 0 & \frac{\Omega}{2} & 0 & 0 \\ \frac{\Omega}{2} & 0 & 0 & \frac{\Omega}{2} & 0 & 0 & \frac{\Omega}{2} & 0 \\ 0 & \frac{\Omega}{2} & \frac{\Omega}{2} & V & 0 & 0 & 0 & \frac{\Omega}{2} \\ \frac{\Omega}{2} & 0 & 0 & 0 & 0 & \frac{\Omega}{2} & \frac{\Omega}{2} & 0 \\ 0 & \frac{\Omega}{2} & 0 & 0 & \frac{\Omega}{2} & V & 0 & \frac{\Omega}{2} \\ 0 & 0 & \frac{\Omega}{2} & 0 & \frac{\Omega}{2} & 0 & V & \frac{\Omega}{2} \\ 0 & 0 & 0 & \frac{\Omega}{2} & 0 & \frac{\Omega}{2} & \frac{\Omega}{2} & 3V \end{pmatrix}$$

c)

d) Solution

$$H_{eff} = \frac{\Omega}{2} \sum_{i=1}^3 \sigma_x^{(i)}$$

Act H_{eff} on $|000\rangle$

$$H_{eff}|000\rangle = \frac{\Omega}{2} (\sigma_x^{(1)}|000\rangle + \sigma_x^{(2)}|000\rangle + \sigma_x^{(3)}|000\rangle)$$

$$= \frac{\Omega}{2} (|100\rangle + |010\rangle + |001\rangle)$$

$$= \frac{\Omega}{2} \left(\frac{1}{\sqrt{3}} (|100\rangle + |010\rangle + |001\rangle) \right)$$

$$= \frac{\Omega}{2} \left(\underbrace{\frac{|100\rangle + |010\rangle + |001\rangle}{\sqrt{3}}}_{=|W\rangle} \right)$$

$$= \frac{\Omega}{2} |W\rangle$$

~~///~~

$$= \frac{\Omega}{2} |W\rangle$$

~~///~~

Act H_{eff} on $|W\rangle$ (and ignore 2-excitation state)

$$H_{eff}|W\rangle = H_{eff} \left(\frac{|100\rangle + |010\rangle + |001\rangle}{\sqrt{3}} \right)$$

$$= \frac{1}{\sqrt{3}} (H_{eff}|100\rangle + H_{eff}|010\rangle + H_{eff}|001\rangle)$$

$$= \frac{1}{\sqrt{3}} \left(\frac{\Omega}{2} (\sigma_x^{(1)}|100\rangle + \sigma_x^{(2)}|010\rangle + \sigma_x^{(3)}|001\rangle) \right)$$

$$= \frac{\Omega}{2\sqrt{3}} [|110\rangle + |101\rangle + |100\rangle + |100\rangle + |100\rangle + |011\rangle + |000\rangle + |110\rangle + |101\rangle]$$

$$= \frac{\sqrt{3}}{2} \Omega \left(3|1000\rangle + 2|1011\rangle + 2|1101\rangle + 2|1110\rangle \right)$$

Note that: - $|1000\rangle$ appears 3 times
- 2-excitation states (Blocked, excluded)

$$H_{\text{eff}}|100\rangle \approx \frac{\sqrt{3}}{2} \Omega |1000\rangle$$

$$\approx \frac{\sqrt{3}}{2} \Omega |1000\rangle$$

① Therefore

$$\therefore \left. \begin{array}{l} H_{\text{eff}}|1000\rangle \approx |100\rangle \\ H_{\text{eff}}|100\rangle \approx |1000\rangle \end{array} \right\}$$

② The evolution never leaves

span of $|1000\rangle, |100\rangle$

c) Solution

$$\text{Span} = \{ |1000\rangle, |100\rangle \}$$

$$|100\rangle = \frac{|1001\rangle + |1010\rangle + |1100\rangle}{\sqrt{3}}$$

Effective Hamiltonian in $\{ |1000\rangle, |100\rangle \}$

$$H|1000\rangle = \frac{\sqrt{3}}{2} \Omega |100\rangle$$

$$H|100\rangle = \frac{\sqrt{3}}{2} \Omega |1000\rangle$$

$$H_{\text{eff}} = \frac{\sqrt{3}}{2} \Omega \sigma_x$$

$$= \begin{pmatrix} 0 & \frac{\sqrt{3}}{2} \Omega \\ \frac{\sqrt{3}}{2} \Omega & 0 \end{pmatrix}$$

//

Time evolution and then $|1000\rangle$

$$|4\rangle = |1000\rangle$$

$$H = g \sigma_x = \frac{\sqrt{3}}{2} \Omega \sigma_x$$

$$|\psi(t)\rangle = \cos(gt) |1000\rangle - i \sin(gt) |V0\rangle \quad \text{--- Quantum state}$$

The population of state, $|4\rangle = |1000\rangle$

$$\begin{aligned} \textcircled{1} P_{1000}(t) &= |\langle 1000 | \psi(t) \rangle|^2 = \cos^2(gt) = \cos^2\left(\frac{\sqrt{3}}{2} \Omega t\right) \\ &= \cos^2\left(\frac{\sqrt{3}}{2} \Omega t\right) \end{aligned}$$

When, the probability become zero

$$\boxed{P_{1000}(t) = 0 \rightarrow \text{(Entry deleted) mean}}$$

$$\cos^2\left(\frac{\sqrt{3}}{2} \Omega t\right) = 0$$

$$\cos\left(\frac{\sqrt{3}}{2} \Omega t\right) = 0$$

$$\frac{\sqrt{3}}{2} \Omega t = \frac{\pi}{2} + k\pi, \quad k = 0, 1, 2, \dots$$

$$\frac{\sqrt{3} \Omega t}{2} = \frac{\pi}{2} + k\pi$$

$$t = \frac{2}{\sqrt{3} \Omega} \left(\frac{\pi}{2} + k\pi \right)$$

$$= \frac{2}{\sqrt{3} \Omega} \left(\frac{\pi + 2k\pi}{2} \right)$$

$$= \frac{2}{\sqrt{3} \Omega} \cdot \frac{1}{2} (\pi + 2k\pi) = \frac{\pi}{\sqrt{3} \Omega} (1 + 2k)$$

$$= \frac{\pi}{\sqrt{3} \Omega} (1 + 2k), \quad k = 0, 1, 2, \dots$$

$$\underline{\underline{\pi}}$$

The earliest time ($k=0$)

$$t = \frac{\pi}{\sqrt{3} \Omega} //$$

$$) \quad t = \underline{\underline{\frac{\pi}{\sqrt{3} \Omega}}}$$

4) solution

~~9(1) 1st attempt~~

$$S_{23}^{(H)} = \text{Tr}_2(|W\rangle\langle W|)$$

$$= \text{Tr}_2 \left[\frac{1}{13} (|001\rangle + |010\rangle + |100\rangle) \frac{1}{13} (|001\rangle + |010\rangle + |100\rangle) \right]$$

$$= \frac{1}{3} \text{Tr}_2 \left[|001\rangle\langle 001| + |001\rangle\langle 010| + |001\rangle\langle 100| + |010\rangle\langle 001| + |010\rangle\langle 010| + |010\rangle\langle 100| \right. \\ \left. + |100\rangle\langle 001| + |100\rangle\langle 010| + |100\rangle\langle 100| \right]$$

$$= \frac{1}{3} \left[|01\rangle\langle 01| + |01\rangle\langle 10| + 0 + |10\rangle\langle 01| + |10\rangle\langle 10| + |00\rangle\langle 00| \right]$$

$$= \frac{1}{3} \left[\underbrace{|01\rangle\langle 01| + |01\rangle\langle 10| + |10\rangle\langle 01| + |10\rangle\langle 10|}_{= 2|\psi^+\rangle\langle\psi^+|} + |00\rangle\langle 00| \right]$$

$$= \frac{2}{3} |\psi^+\rangle\langle\psi^+| + \frac{1}{3} |00\rangle\langle 00|$$

$$S_{23}^{(H)} = \frac{2}{3} |\psi^+\rangle\langle\psi^+| + \frac{1}{3} |00\rangle\langle 00|$$

$$(S_{23}^{(H)})^2 = \left(\frac{2}{3} |\psi^+\rangle\langle\psi^+| + \frac{1}{3} |00\rangle\langle 00| \right) \left(\frac{2}{3} |\psi^+\rangle\langle\psi^+| + \frac{1}{3} |00\rangle\langle 00| \right)$$

$$= \frac{4}{9} |\psi^+\rangle\langle\psi^+| \underbrace{|\psi^+\rangle\langle\psi^+|}_{=1} + \frac{1}{9} \underbrace{|00\rangle\langle 00|00\rangle\langle 00|}_{=1}$$

$$= \frac{4}{9} |\psi^+\rangle\langle\psi^+| + \frac{1}{9} |00\rangle\langle 00|$$

$$\text{purity} = \text{Tr}((S_{23}^{(H)})^2) = \text{Tr} \left[\frac{4}{9} |\psi^+\rangle\langle\psi^+| + \frac{1}{9} |00\rangle\langle 00| \right]$$

$$= \frac{4}{9} \underbrace{\text{Tr}(|\psi^+\rangle\langle\psi^+|)}_{=1} + \frac{1}{9} \underbrace{\text{Tr}(|00\rangle\langle 00|)}_{=1}$$

$$= \frac{4}{9} + \frac{1}{9} = \frac{4+1}{9} = \frac{5}{9}$$

Solution

$$|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$$

$$S_{GHZ} = |GHZ\rangle\langle GHZ|$$

$$= \frac{1}{2} (|000\rangle\langle 000| + |000\rangle\langle 111| + |111\rangle\langle 000| + |111\rangle\langle 111|)$$

$$S_{H2} = \text{Tr}_G(S_{GHZ})$$

$$= \frac{1}{2} (|00\rangle\langle 00| + |11\rangle\langle 11|)$$

$$S_{H2}^2 = \frac{1}{2} (|00\rangle\langle 00| + |11\rangle\langle 11|) \left(\frac{1}{2} (|00\rangle\langle 00| + |11\rangle\langle 11|) \right)$$

$$= \frac{1}{4} (|00\rangle\langle 00| + |11\rangle\langle 11|)$$

$$\text{Purity} = \text{Tr}(S_{H2}^2)$$

$$= \text{Tr}\left(\frac{1}{4} (|00\rangle\langle 00| + |11\rangle\langle 11|)\right)$$

$$= \frac{1}{4} [\text{Tr}(|00\rangle\langle 00|) + \text{Tr}(|11\rangle\langle 11|)]$$

$$= \frac{1}{4} (2 + 2) = \frac{2}{4} = \underline{\underline{\frac{1}{2}}}$$

$$= \underline{\underline{\frac{1}{2}}}$$

Purity of GHZ and W case

$$\text{Purity}(GHZ) < \text{Purity}(W)$$

$$\underline{\underline{\frac{1}{2}}} < \underline{\underline{\frac{5}{9}}} \Rightarrow \underline{\underline{\frac{1}{2}}} < \underline{\underline{\frac{5}{9}}}$$

more mixing
 $\text{Purity}(GHZ) = \frac{1}{2}$
 $\text{Purity}(S_{H2}) = \frac{5}{9}$
 less mixing

5) From another paper (problem-5) where

- Show that the Wener state of two

$$S(d,a) = I \otimes I + aU/d^2 + ad$$

where
 $d = 4$
 $a = -\frac{1}{2}$ (9)

$$S(a) = \frac{I \otimes I + aU}{d^2 + ad} = \frac{I + aU}{d^2 + ad} = \frac{I + (-\frac{1}{2})U}{4^2 + (-\frac{1}{2})4} = \frac{I - \frac{1}{2}U}{16-2}$$

$$= \frac{I - \frac{1}{2}U}{14} = \frac{1}{14} (I - \frac{1}{2}U)$$

$$S^T = \frac{1}{14} (I - \frac{1}{2}U^T), \quad U^T = |5272521$$

$$S^T = \frac{1}{14} (I - \frac{1}{2}|527454\rangle), \quad \text{where}$$

$$= 4|d+74d+1\rangle$$

$$|d+7 = \frac{1}{14}|527$$

$$|5274521 = d|d+74d+1$$

$$|5274521 = 4|d+74d+1$$

$$S^T = \frac{1}{14} (I - \frac{1}{2} \cdot 4 |d+74d+1\rangle)$$

$$= \frac{1}{14} (I - \frac{4}{2} |d+74d+1\rangle)$$

$$= \frac{1}{14} (I - 2 |d+74d+1\rangle)$$

Define the operator

$$U = I - 2 |d+74d+1\rangle$$

$$S^T = \frac{1}{14} (U)$$

$$= \frac{U}{14}$$

Check U is dichotomic unitary

$$U^2 = (I - 2P)^2, \quad \text{where, } P = |d+74d+1\rangle$$

$$= I - 4P + 4P^2$$

$$P^2 = P \text{ (projector)}$$

$$= I - 4P + 4P$$

$$U^2 = I$$

Two copies $(ST)^{\otimes 2}$

$$(ST)^{\otimes 2} = \left(\frac{1}{14} U\right) \left(\frac{1}{14} U\right) = \frac{1}{196} (U \otimes U) = \frac{1}{196} (U \otimes U)$$

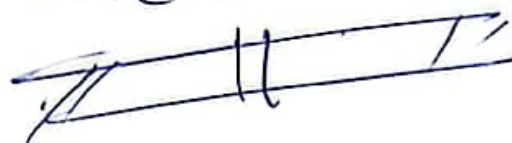
convert "2-copy distinguishable" into a concave function

$$\langle \psi | (ST)^{\otimes 2} | \psi \rangle < 0$$

$$\langle \psi | (ST)^{\otimes 2} | \psi \rangle = \langle \psi | \frac{1}{196} (U \otimes U) | \psi \rangle$$

$$= \frac{1}{196} \langle \psi | (U \otimes U) | \psi \rangle$$

$$= \frac{1}{196} \langle \psi | (U \otimes U) | \psi \rangle$$



where,

$$\frac{1}{196} > 0$$

Distinguishable (2-copy)

$$- \langle \psi | (U \otimes U) | \psi \rangle < 0$$

Non-distinguishable (2-copy)

$$\langle \psi | (U \otimes U) | \psi \rangle \geq 0$$

The equivalent "Kronecker sum"

$$TV(A+A) + TV(B+B) = \frac{1}{A}$$

$$A \oplus B = A \oplus I + I \oplus B$$

Let, $s_1 \geq s_2 \geq \dots$ be the singular values of

$A \oplus B$. Then prove

$$S_1^2 + S_2^2 = \frac{1}{2}$$

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